

Notes.

(a) You may freely use any result proved in class unless you have been asked to prove the same. Use your judgement. All other steps must be justified.

(b) We use $\mathbb{R}$ = real numbers; $I = [0, 1]$, the unit interval in $\mathbb{R}$; S^n = the unit sphere in $\mathbb{R}^{n+1}$.

1. [20 Points] Let X be a topological space.

- (i) For any point $x_0 \in X$, define the fundamental group $\pi_1(X, x_0)$ first as a set together with a multiplication law in it. Explain why the multiplication law is well-defined. Give the identity element and describe the inverse for any element in $\pi_1(X, x_0)$. (You need not check the associativity property).
- (ii) Prove that for any continuous map $h: (Y, y_0) \rightarrow (Z, z_0)$ of pointed spaces, there is a well-defined function $h_*: \pi_1(Y, y_0) \rightarrow \pi_1(Z, z_0)$ and verify that it is in fact a group homomorphism. Prove that if h is a homeomorphism, then h_* is an isomorphism.

2. [20 Points] Let $f: (E, e_0) \rightarrow (X, x_0)$ be a continuous map of pointed topological spaces.

- (i) Define what it means for f to be a covering map.
- (ii) Prove that if f is a covering map, then $f_*: \pi_1(E, e_0) \rightarrow \pi_1(X, x_0)$ is injective. (You may use any lifting property of covering maps proved in class, but make sure to state what precisely you are using). Give an example (without proof) where f_* is not surjective.

3. [40 Points] Define the projective space $\mathbb{R}P^2$.

- (i) Give a covering space for $\mathbb{R}P^2$ (you must check that the map you specify is a covering map) and using this, compute the fundamental group of $\mathbb{R}P^2$.
- (ii) Construct a simplicial complex K whose associated polytope $|K|$ is homeomorphic to $\mathbb{R}P^2$. Compute its homology groups.

4. [20 Points] Let K be a simplicial complex.

- (i) For any integer p , define the group $C_p(K)$ of oriented p -chains on K .
- (ii) For any integer p , define the boundary map $\partial_p: C_p(K) \rightarrow C_{p-1}(K)$. Check that $\partial_p \circ \partial_{p+1} = 0$.